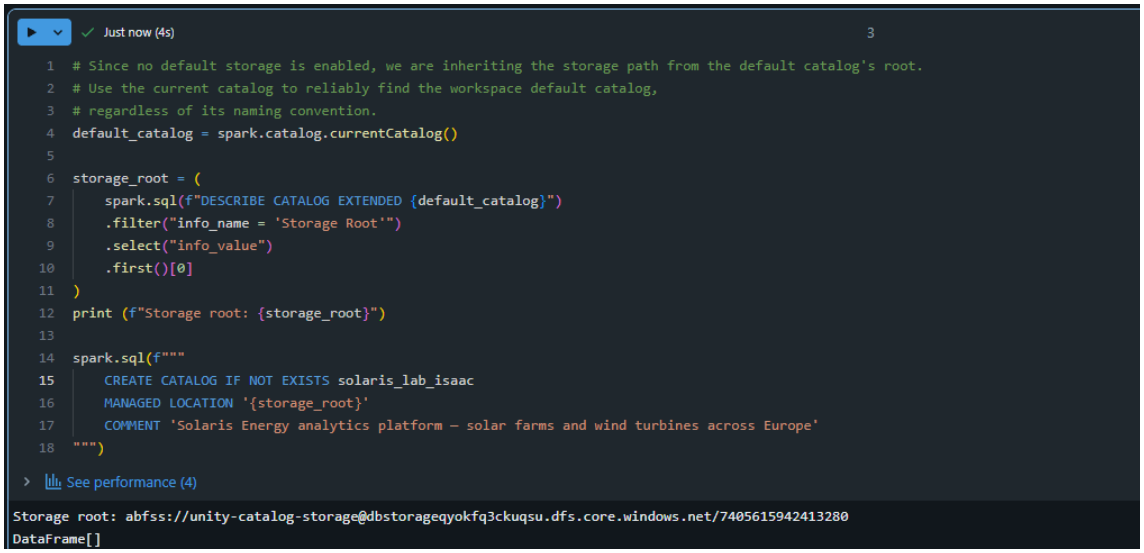
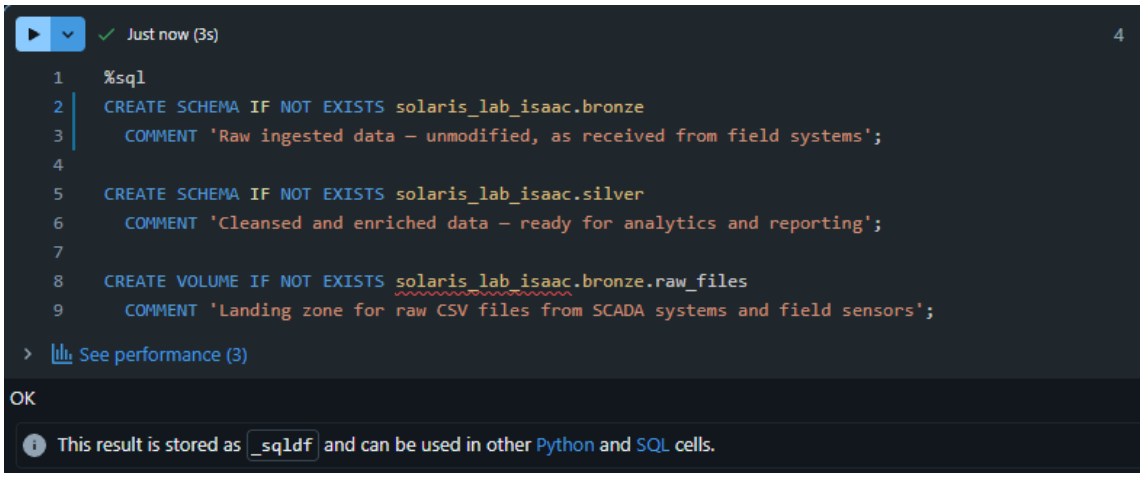
Contenido

Ejercicio 1 2
Task 1 2
Ejercicio 2 3
Task 2.1 3
Ejercicio 2.3 4
Ejercicio 2.4 4
Ejercicio 3 5
Ejercicio 3.1 5
Ejercicio 3.4 7
Ejercicio 4 7
Ejercicio 4.1 8
Ejercicio 4.2 9
Ejercicio 4.3 9
Limpieza 9

Ejercicio 1

Task 1





Llenar el volumen

```
EVT009,T-DK-004,WIND_DK_001,2025-06-01 10:33:00,Maintenance,Low,Tower bolt torque check completed,true
EVT010,T-DK-005,WIND_DK_001,2025-06-01 14:18:00,Alert,Medium,Ice accretion on blades - de-icing initiated,true
EVT011,T-NL-001,WIND_NL_001,2025-06-02 05:30:00,Maintenance,Low,Hydraulic fluid top-up completed,true
EVT012,T-NL-002,WIND_NL_001,2025-06-02 08:00:00,Alert,High,Generator winding temperature 12 degrees C above limit,false
EVT013,T-NL-003,WIND_NL_001,2025-06-02 09:45:00,Fault,Medium,Anemometer replaced after sensor malfunction,true
EVT014,T-NL-004,WIND_NL_001,2025-06-02 12:10:00,Inspection,Low,Control system firmware update applied,true
EVT015,T-NL-005,WIND_NL_001,2025-06-02 15:22:00,Alert,Low,Brake pad wear indicator triggered,false
EVT016,T-DK-001,WIND_DK_001,2025-06-02 03:18:00,Maintenance,Medium,Main bearing grease replenishment,true
EVT017,T-DK-002,WIND_DK_001,2025-06-02 07:55:00,Alert,High,Pitch battery backup low - replacement scheduled,false
EVT018,T-DK-003,WIND_DK_001,2025-06-02 10:40:00,Fault,Critical,Transformer fault - turbine offline for 4 hours,true
EVT019,T-DK-004,WIND_DK_001,2025-06-02 13:15:00,Inspection,Low,Cooling system filters cleaned,true
EVT020,T-DK-005,WIND_DK_001,2025-06-02 16:50:00,Maintenance,Medium,Slip ring inspection and cleaning completed,true
"""

with open("/Volumes/solaris_lab_isaac/bronze/raw_files/solar_readings.csv", "w") as f:
    f.write(solar_csv.strip())

with open("/Volumes/solaris_lab_isaac/bronze/raw_files/turbine_events.csv", "w") as f:
    f.write(turbine_csv.strip())

print("✅ Sample data written to /Volumes/solaris_lab_isaac/bronze/raw_files/")

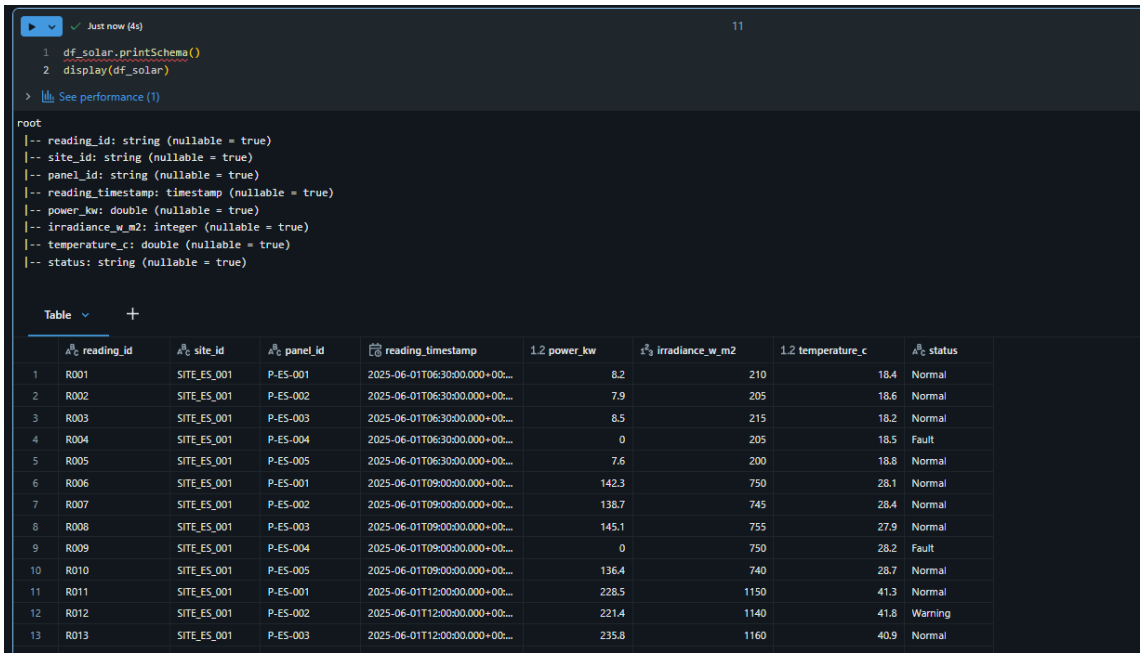
✅ Sample data written to /Volumes/solaris_lab_isaac/bronze/raw_files/
```

Ejercicio 2

Task 2.1



Task 2.2



Ejercicio 2.3

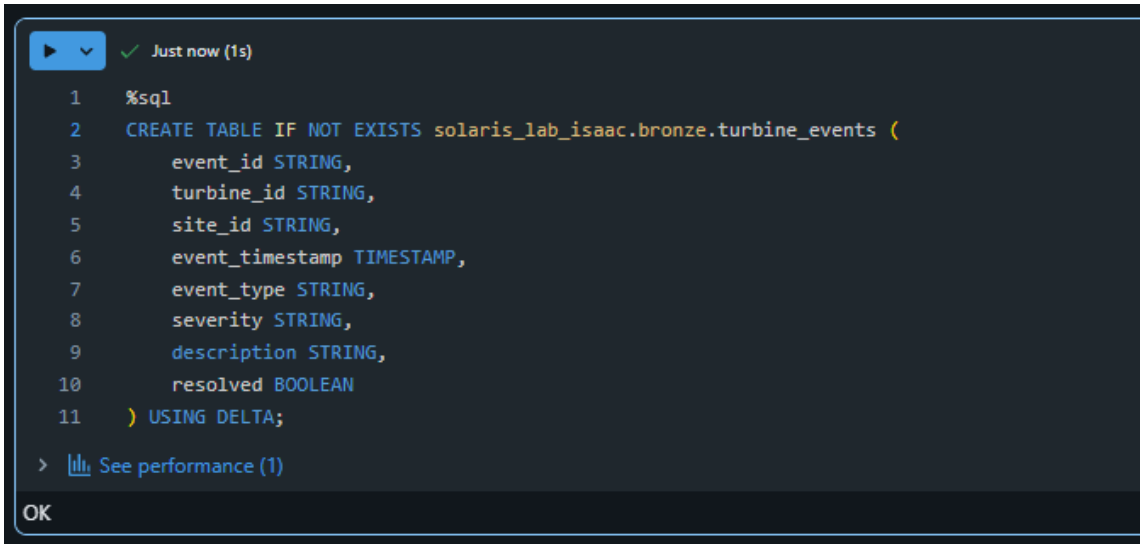


Ejercicio 2.4

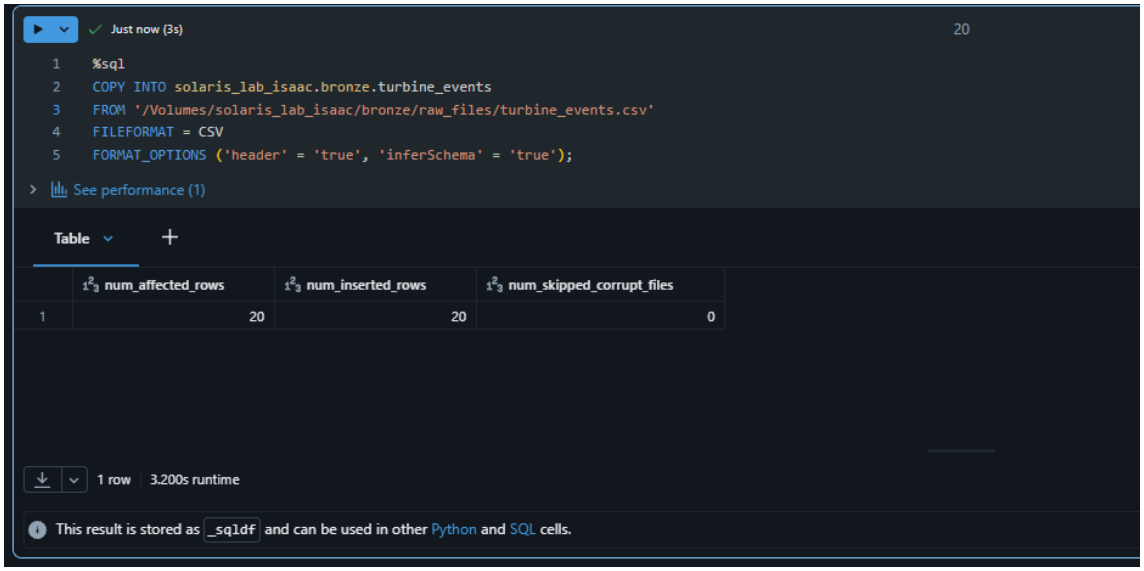


Ejercicio 3

Ejercicio 3.1



Ejercicio 3.2



Ejercicio 3.3

Just now (1s)

```
%sql
SELECT COUNT(*) FROM solaris_lab_isaac.bronze.turbine_events;
```

See performance (1)

Table

	COUNT(*)
1	20

1 row | 1.347s runtime

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

Just now (2s)

```
%sql
COPY INTO solaris_lab_isaac.bronze.turbine_events
FROM '/Volumes/solaris_lab_isaac/bronze/raw_files/turbine_events.csv'
FILEFORMAT = CSV
FORMAT_OPTIONS ('header' = 'true', 'inferSchema' = 'true');
```

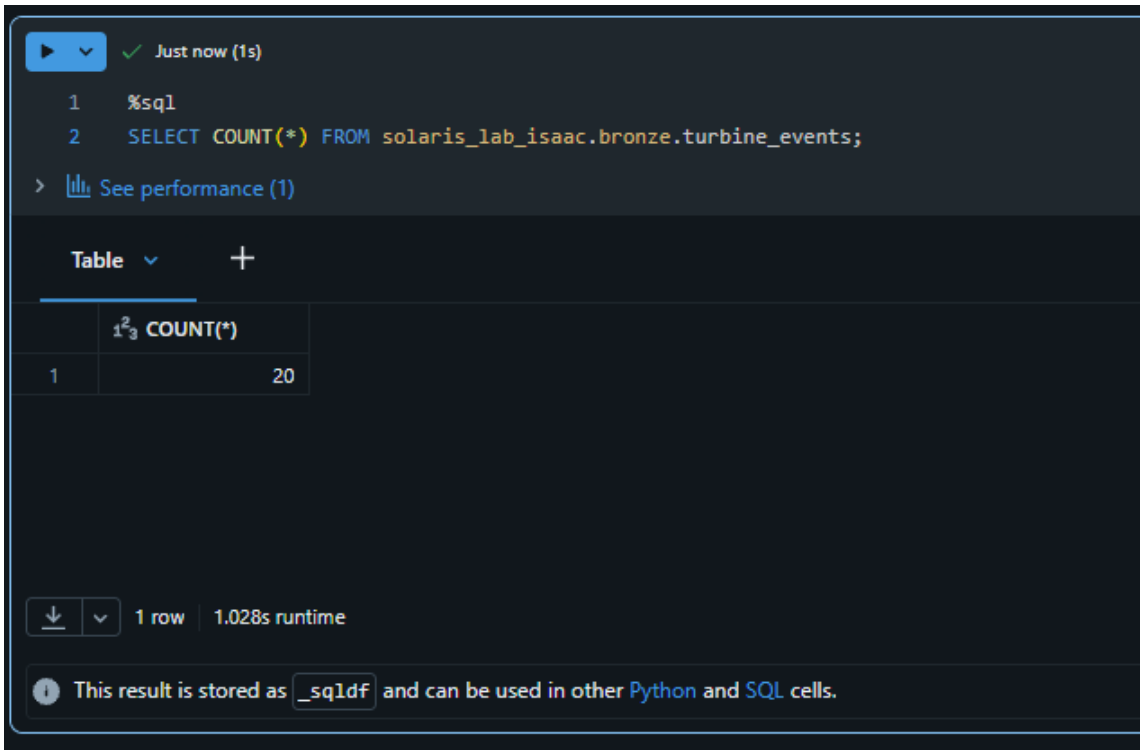
See performance (1)

Table

	num_affected_rows	num_inserted_rows	num_skipped_corrupt_files
1	0	0	0

1 row | 2.053s runtime

This result is stored as `_sqldf` and can be used in other Python and SQL cells.

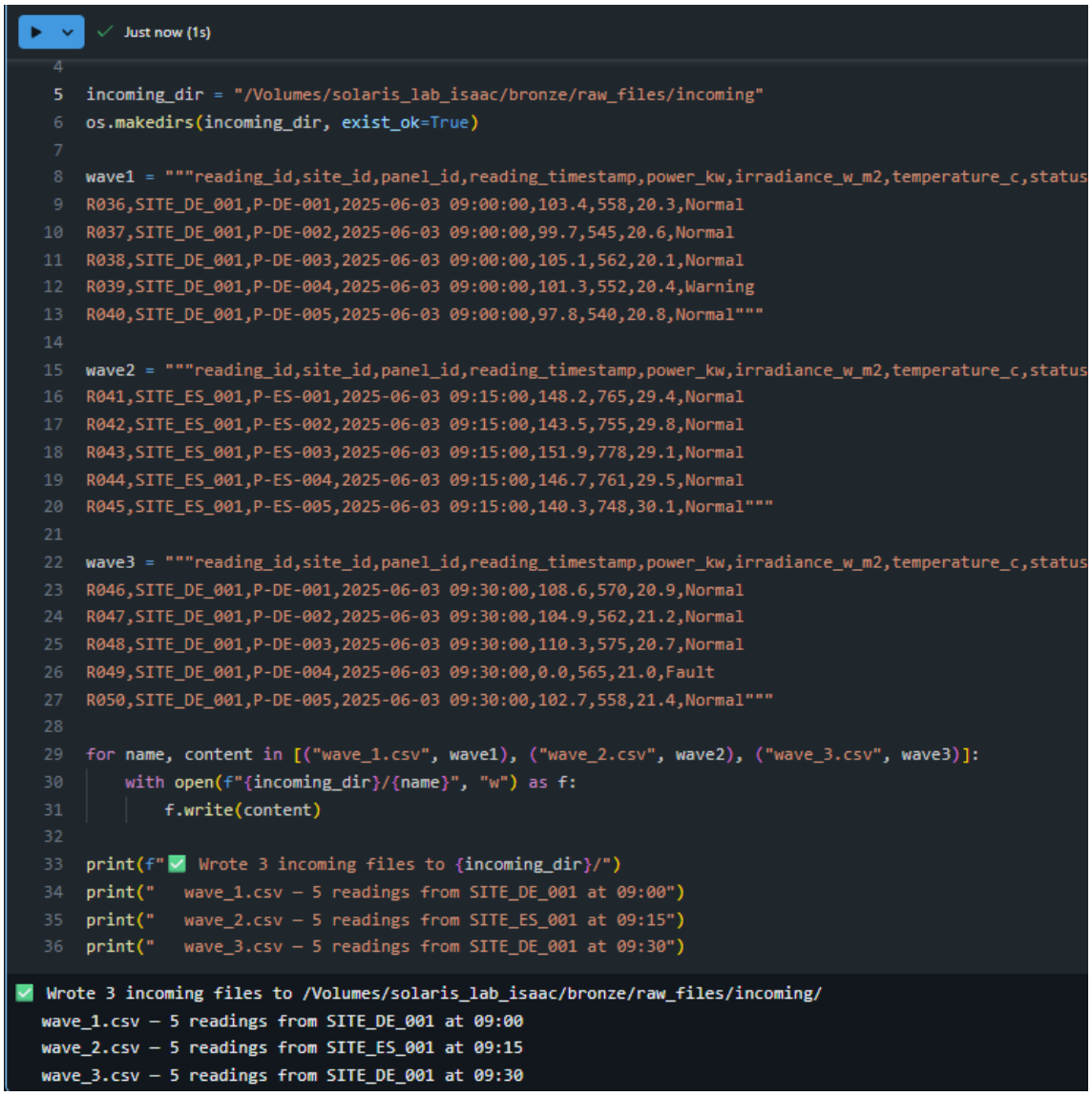


Ejercicio 3.4



Ejercicio 4

Añadir los datos



```
Just now (1s)

incoming_dir = "/Volumes/solaris_lab_isaac/bronze/raw_files/incoming"
os.makedirs(incoming_dir, exist_ok=True)

wave1 = """reading_id,site_id,panel_id,reading_timestamp,power_kw,irradiance_w_m2,temperature_c,status
R036,SITE_DE_001,P-DE-001,2025-06-03 09:00:00,103.4,558,20.3,Normal
R037,SITE_DE_001,P-DE-002,2025-06-03 09:00:00,99.7,545,20.6,Normal
R038,SITE_DE_001,P-DE-003,2025-06-03 09:00:00,105.1,562,20.1,Normal
R039,SITE_DE_001,P-DE-004,2025-06-03 09:00:00,101.3,552,20.4,Warning
R040,SITE_DE_001,P-DE-005,2025-06-03 09:00:00,97.8,540,20.8,Normal"""

wave2 = """reading_id,site_id,panel_id,reading_timestamp,power_kw,irradiance_w_m2,temperature_c,status
R041,SITE_ES_001,P-ES-001,2025-06-03 09:15:00,148.2,765,29.4,Normal
R042,SITE_ES_001,P-ES-002,2025-06-03 09:15:00,143.5,755,29.8,Normal
R043,SITE_ES_001,P-ES-003,2025-06-03 09:15:00,151.9,778,29.1,Normal
R044,SITE_ES_001,P-ES-004,2025-06-03 09:15:00,146.7,761,29.5,Normal
R045,SITE_ES_001,P-ES-005,2025-06-03 09:15:00,140.3,748,30.1,Normal"""

wave3 = """reading_id,site_id,panel_id,reading_timestamp,power_kw,irradiance_w_m2,temperature_c,status
R046,SITE_DE_001,P-DE-001,2025-06-03 09:30:00,108.6,570,20.9,Normal
R047,SITE_DE_001,P-DE-002,2025-06-03 09:30:00,104.9,562,21.2,Normal
R048,SITE_DE_001,P-DE-003,2025-06-03 09:30:00,110.3,575,20.7,Normal
R049,SITE_DE_001,P-DE-004,2025-06-03 09:30:00,0.0,565,21.0,Fault
R050,SITE_DE_001,P-DE-005,2025-06-03 09:30:00,102.7,558,21.4,Normal"""

for name, content in [("wave_1.csv", wave1), ("wave_2.csv", wave2), ("wave_3.csv", wave3)]:
    with open(f"{incoming_dir}/{name}", "w") as f:
        f.write(content)

print(f"✅ Wrote 3 incoming files to {incoming_dir}/")
print("   wave_1.csv — 5 readings from SITE_DE_001 at 09:00")
print("   wave_2.csv — 5 readings from SITE_ES_001 at 09:15")
print("   wave_3.csv — 5 readings from SITE_DE_001 at 09:30")

✅ Wrote 3 incoming files to /Volumes/solaris_lab_isaac/bronze/raw_files/incoming/
wave_1.csv — 5 readings from SITE_DE_001 at 09:00
wave_2.csv — 5 readings from SITE_ES_001 at 09:15
wave_3.csv — 5 readings from SITE_DE_001 at 09:30
```

Ejercicio 4.1

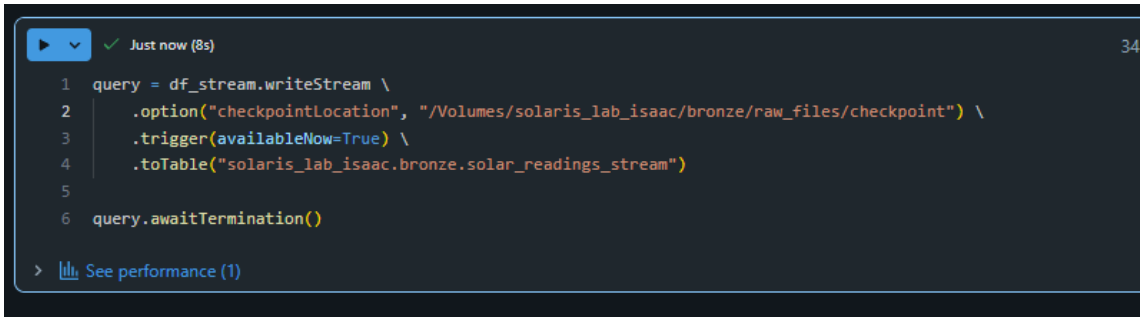


```
Just now (2s)                                                                    32

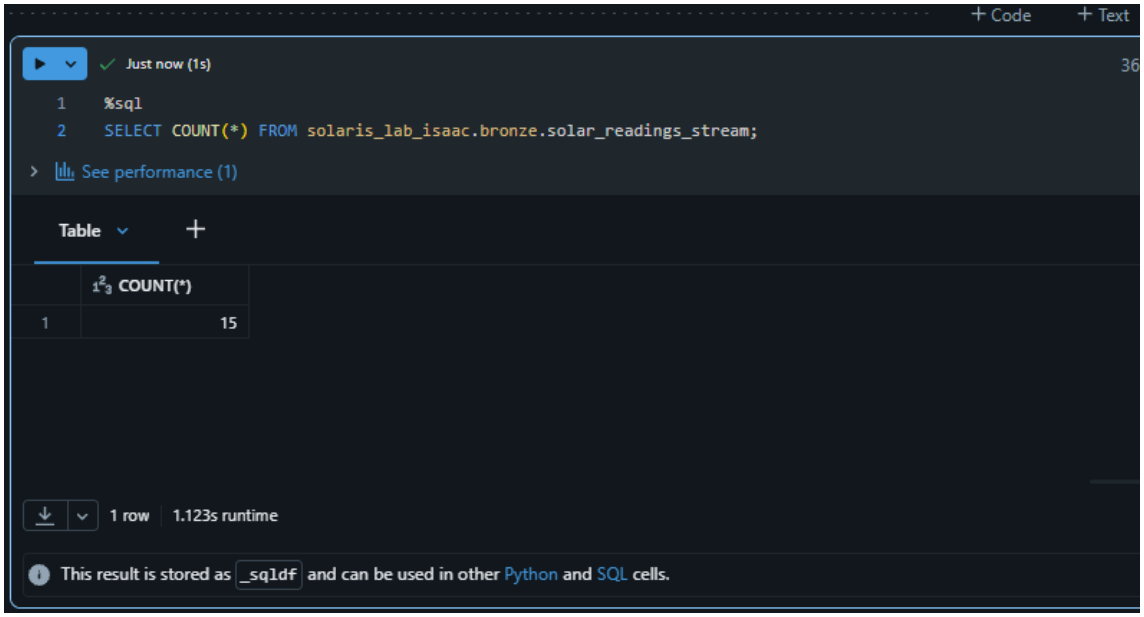
df_stream = spark.readStream \
    .format("cloudFiles") \
    .option("cloudFiles.format", "csv") \
    .option("cloudFiles.schemaLocation", "/Volumes/solaris_lab_isaac/bronze/raw_files/schema") \
    .option("cloudFiles.inferColumnTypes", "true") \
    .option("header", "true") \
    .load("/Volumes/solaris_lab_isaac/bronze/raw_files/incoming/")

df_stream: pyspark.sql.connect.dataframe.DataFrame = [reading_id: string, site_id: string ... 7 more fields]
```

Ejercicio 4.2



Ejercicio 4.3



Limpieza

